

Supply Chain Management Analytics

Executive Summary:

This project analyzes supply chain performance across inventory, demand, suppliers, warehouses, and risk factors. The goal is to identify inefficiencies, reduce stockouts, optimize inventory levels, and improve supplier reliability.

Key outcomes:

- Identified **stock imbalance issues** (overstock + stockouts)
- Highlighted **high-risk suppliers and delayed deliveries**
- Found **demand-supply mismatches across categories**
- Provided **data-driven reorder and optimization strategies**

Inventory Insights:

- A significant portion of SKUs fall under **low stock or out-of-stock categories**, indicating poor replenishment planning.
- Some SKUs show **excess inventory**, leading to holding costs.
- **Inventory Coverage Gap** shows many products cannot meet demand during lead time.

Insight: Inventory is **not balanced** — simultaneous overstocking and understocking.

Demand Analysis:

- Demand distribution is **right-skewed**:
 - Most products have low demand
 - Few products drive high demand (fast-moving SKUs)
- High-demand SKUs are **more prone to stockouts**
- Focus should be on **ABC classification**:
 - A → High priority
 - B → Moderate
 - C → Low

Supplier Performance:

- Majority of suppliers fall under **Class A (Good)**, but a small portion shows delays.
- Some suppliers are close to **risk threshold**, requiring monitoring.
- Supplier On-Time % varies significantly.

Insight: Even small delays impact high-demand products → causes stockouts

Warehouse Analysis:

- Uneven stock distribution across warehouses
- Some warehouses:
 - Overstocked
 - Others facing shortages

Insight: Lack of **inventory redistribution strategy**

Risk Analysis:

- High **stockout risk** observed in fast-moving categories
- Supplier delays + low stock = **critical risk zones**
- Risk Score identifies vulnerable SKUs

Insight: Risk is concentrated in:

- High demand + low stock + long lead time

Reorder & Optimization Insights:

- Many SKUs flagged for reorder
- Some reorder decisions are delayed
- Optimization score highlights inefficiencies

Insight: Reorder system needs:

- Automation
- Threshold-based alerts

Critical Business Insight (Highlight)

CRITICAL: Demand significantly exceeds available stock in key categories. Pantry category shows extreme mismatch between forecast and inventory. Immediate replenishment and supplier coordination required to avoid stockouts.

OPTIMIZATION:

Inventory:

- Implement **safety stock levels**
- Use **demand forecasting models**

Supplier:

- Monitor suppliers near risk threshold
- Introduce **SLA agreements**

Warehouse:

- Redistribute stock across warehouses
- Use **real-time inventory tracking**

Reorder Strategy:

- Automate reorder flags
- Use:

Reorder if $\text{Stock} < \text{Demand during Lead Time}$

Risk Reduction:

- Focus on high-risk SKUs
- Maintain buffer stock for critical items

Conclusion:

The analysis reveals that inefficiencies in inventory planning, supplier reliability, and demand forecasting are the key challenges. By implementing optimized reorder strategies, supplier monitoring, and warehouse balancing, the organization can significantly improve operational efficiency and reduce risk.